

$\nu_1 = 1$   
 $\nu_2$

$2\nu_3$   
 $2\nu_4$

$= 3$   
 $= 5$   
 $= 3$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} \nu_2 \\ \nu_3 \\ \nu_4 \\ \nu_5 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 3 \\ 0 \end{pmatrix}$$



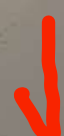
$2\nu_2 + \nu_3 + 2\nu_5 = 0$

$$\div 2 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} \nu_2 \\ \nu_3 \\ \nu_4 \\ \nu_5 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 3 \\ 7 \end{pmatrix}$$

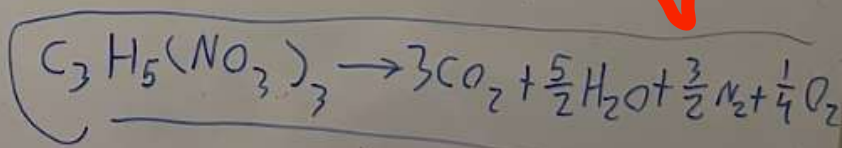
$\div 2$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} \nu_2 \\ \nu_3 \\ \nu_4 \\ \nu_5 \end{pmatrix} = \begin{pmatrix} 3 \\ 5/2 \\ 3 \\ 1/2 \end{pmatrix}$$

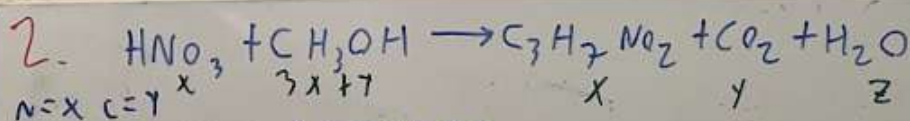
$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \nu_2 \\ \nu_3 \\ \nu_4 \\ \nu_5 \end{pmatrix} = \begin{pmatrix} 3 \\ 5/2 \\ 7/2 \\ 1/4 \end{pmatrix}$$



$$\div 2 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} \nu_2 \\ \nu_3 \\ \nu_4 \\ \nu_5 \end{pmatrix} = \begin{pmatrix} 3 \\ 5/2 \\ 3 \\ 7 \end{pmatrix}$$



P1: 15/15



$$\text{H: } x + 4(3x+7) = 7x + 2z$$

$$\text{O: } (3x+3x+7) = (2x+2y+z) \cdot 2$$

$$\text{H: } 13x + 4y = 7x + 2z$$

$$\text{O: } 12x + 2y = 4x + 4y + 2z$$

$$x + 2y = 3x - 4y$$

$$6y = 2x \rightarrow 3y = x$$

$$3y + 3(3y) + y = 3y + y + z$$

$$13(3y) + 4y = 7(3y) + 2z = \text{H}$$

$$39y + 4y = 21y + 2z$$

$$22y = 2z \rightarrow 11y = z$$

$$\rightarrow 3y + 3(3y) + y = 3y + y + 11y$$

$$3y + 10y = 3y + y + 11y$$



$$54 \frac{\text{mol}}{\text{L}} - 10 \frac{\text{mol}}{\text{L}} = 44 \frac{\text{mol}}{\text{L}} \text{HNO}_3 \rightarrow 0.0449 \text{ L}$$

$$\frac{0.0449}{2} \text{HNO}_3$$

$$\cdot \frac{1 \text{ mol HNO}_3}{63.019 \text{ g HNO}_3}$$

$$\cdot 10 \text{ L} = 0.00698 \text{ mol HNO}_3$$

$$\frac{0.00698}{1 \text{ mol HNO}_3}$$

$$\cdot \frac{10 \text{ mol CH}_3\text{OH}}{3 \text{ mol HNO}_3}$$

$$\cdot \frac{320.43 \text{ g CH}_3\text{OH}}{1 \text{ mol CH}_3\text{OH}}$$

$$= 0.7469 \text{ g CH}_3\text{OH}$$

P2: 20/20

$$3. \quad M_{C_2H_5OH} = (12.01 \cdot 2) + 6(1.008) + 16 = 46.068 \frac{g}{mol}$$

$$M_{CH_4} = 12.01 + 4(1.008) = 16.042 \frac{g}{mol} \quad M_{H_2O} = 16 + 2(1.008) = 18.016 \frac{g}{mol}$$

$$M_{C_6H_{12}O_6} = 6(12.01) + 12(1.008) + 6(16) = 180.156$$

$$C_2H_5OH: \frac{6(1.008)}{46.068} \cdot 100\% = w + 13\% \quad \checkmark$$

$$CH_4: \frac{4(1.008)}{16.042} \cdot 100\% = w + 25\% \quad \checkmark$$

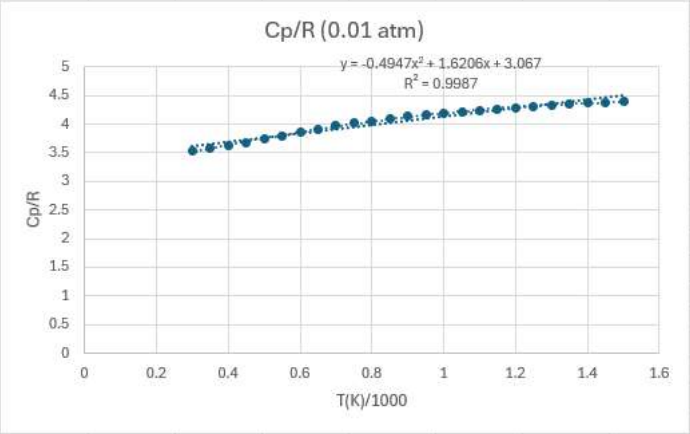
$$H_2O: \frac{2(1.008)}{18.016} \cdot 100\% = w + 11\% \quad \checkmark$$

$$C_6H_{12}O_6: \frac{12(1.008)}{180.156} \cdot 100\% = w + 6.7\% \quad \checkmark$$

Since majority of substance w% exceeds Max of 8% of nanotubes claims are PROBABLY FALSE. ↓

p3: 15/15

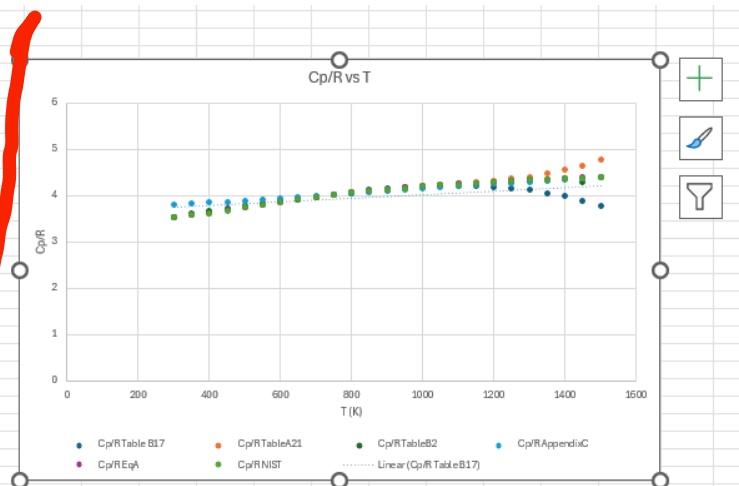
| T(K)/1000 | Cp/R (0.01 atm) |
|-----------|-----------------|
| 0.3       | 3.5345          |
| 0.35      | 3.5717          |
| 0.4       | 3.6212          |
| 0.45      | 3.6787          |
| 0.5       | 3.7396          |
| 0.55      | 3.8008          |
| 0.6       | 3.8599          |
| 0.65      | 3.9155          |
| 0.7       | 3.9672          |
| 0.75      | 4.0145          |
| 0.8       | 4.0577          |
| 0.85      | 4.097           |
| 0.9       | 4.1327          |
| 0.95      | 4.1652          |
| 1         | 4.1948          |
| 1.05      | 4.2219          |
| 1.1       | 4.2469          |
| 1.15      | 4.2698          |
| 1.2       | 4.2912          |
| 1.25      | 4.3112          |
| 1.3       | 4.33            |
| 1.35      | 4.3479          |
| 1.4       | 4.3651          |
| 1.45      | 4.3815          |
| 1.5       | 4.3975          |



#4: 25/125

| T(K) | Cp/R (B17)  | Cp/R (A21) | Cp/R (B: Cp/R (Appendix | Cp/R (eq. 4 | Cp/R (NIST) |
|------|-------------|------------|-------------------------|-------------|-------------|
| 300  | 3.535331489 | 3.5323858  | 3.5372                  | 3.7908      | 3.5345      |
| 350  | 3.583226275 | 3.5752845  | 3.6031                  | 3.8161      | 3.5717      |
| 400  | 3.634896319 | 3.6258976  | 3.6655                  | 3.8414      | 3.6212      |
| 450  | 3.689380894 | 3.6815775  | 3.7246                  | 3.8667      | 3.6787      |
| 500  | 3.745719269 | 3.74       | 3.7805                  | 3.892       | 3.7396      |
| 550  | 3.802950716 | 3.7991645  | 3.8332                  | 3.9173      | 3.8008      |
| 600  | 3.860114506 | 3.8573936  | 3.8829                  | 3.9426      | 3.8599      |
| 650  | 3.91624991  | 3.9133335  | 3.9298                  | 3.9679      | 3.9155      |
| 700  | 3.970306199 | 3.9659536  | 3.9739                  | 3.9932      | 3.9672      |
| 750  | 4.021592645 | 4.0215926  | 4.0153                  | 4.0185      | 4.0145      |
| 800  | 4.068879518 | 4.0587236  | 4.0543                  | 4.0438      | 4.0577      |
| 850  | 4.1123309   | 4.0984415  | 4.0908                  | 4.0691      | 4.097       |
| 900  | 4.141785631 | 4.1339456  | 4.125                   | 4.0944      | 4.1327      |
| 950  | 4.177665414 | 4.1658285  | 4.157                   | 4.1197      | 4.1652      |
| 1000 | 4.199701708 | 4.195      | 4.187                   | 4.145       | 4.1948      |
| 1050 | 4.213027385 | 4.2226935  | 4.2151                  | 4.1703      | 4.2219      |
| 1100 | 4.216670917 | 4.2504656  | 4.2414                  | 4.1956      | 4.2469      |
| 1150 | 4.209682373 | 4.2801965  | 4.2656                  | 4.2209      | 4.2698      |
| 1200 | 4.191097426 | 4.3140896  | 4.2889                  | 4.2462      | 4.2912      |
| 1250 | 4.159955348 | 4.3546719  | 4.3104                  | 4.2715      | 4.3112      |
| 1300 | 4.115295045 | 4.4047936  | 4.3306                  | 4.2968      | 4.33        |
| 1350 | 4.056156874 | 4.4676285  | 4.3495                  | 4.3221      | 4.3479      |
| 1400 | 3.981579023 | 4.5466736  | 4.3673                  | 4.3474      | 4.3651      |
| 1450 | 3.890601125 | 4.6457498  | 4.2842                  | 4.3727      | 4.3815      |
| 1500 | 3.782262449 | 4.769      | 4.4002                  | 4.398       | 4.3975      |

✓





1. When comparing fits for  $C_p/R$  versus  $T$  from NIST to the polynomial fit obtained in part (a), I found that the NIST values are valid over temperatures 300-1500K. These two functions closely match within this interval and the trendlines are almost the same. In assessing the quality of the NIST data, the third order polynomial fit from part (a) yields an  $R^2$  value of 0.9984 which shows a good agreement between fit and data.
2. Comparing  $C_p/R$  versus  $T$  from B17 to the polynomial fit from part (a) it was found that the values represent the data well within the temperature range of 300 to 1150K, but beyond 1200 and up to 1500K, Table B17 doesn't fit the data well. With the fit quality, the third order polynomial fit got an  $R^2$  value of 1 for Table B17 with a perfect correspondence within its valid range.
3. Comparing the fit of  $C_p/R$  versus  $T$  from Table A21 to the polynomial fit found in (a) I found that the values from A21 are accurate within the temperature range of 300 to 1250K but from 1300 to 1500K it doesn't fit well anymore. With the fit quality, the third order polynomial for A21 yields  $R^2$  of 0.9945 showing strong correlation. Since A21 is fourth order increasing the polynomial degree to four improves the fit with  $R^2 = 1$ , making it perfect
4. Comparing  $C_p/R$  versus  $T$  from Appendix C to the polynomial fit from (a) I found that the values in Appendix C are accurate within the range of 650-1500K, but the data doesn't align well with the polynomial fit for the lower temperatures specifically from 300 to 600K. With the fit quality, the third order polynomial for Appendix C has an  $R^2 = 1$  showing a perfect fit.
5. Comparing  $C_p/R$  versus  $T$  from Table B2 to the polynomial fit from (a) I found that Table B2 provides accurate values throughout the 300-1500K range. For the fit quality, the third order polynomial fit for Table B2 has an  $R^2 = 1$  showing a perfect correlation.

NO PS? ORS